Musanze Harvest Intelligence Service

SWE 3513 Artificial Intelligence - Individual Assignment 2 - Technical Report

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Verification code	**AI-A2-3862** (seed 3862)
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Decision problem and stakeholder

A potato cooperative in Musanze receives farm records in inconsistent spreadsheets. Its **field officer** needs an evidence-based tool to support visit planning. The service answers three questions from seven farm descriptors (rainfall, temperature, soil moisture, fertilizer, field size, season, cooperative support):

1. **Regression:** how many tons of potatoes will this farm harvest (yield_tons)?
2. **Classification:** is the farm at high production risk, i.e. is its yield per hectare below the median (high_risk)?
3. **Clustering:** which group of farms with similar conditions does it resemble?

Internet access may be unreliable, so training and command-line prediction run locally; FastAPI is provided for a later mobile app.

Ethical boundary

The dataset is **synthetic** (generated from my registration digits) and the system is a classroom decision-support prototype. It must not replace an agronomist and must never be used to decide access to finance, insurance or public services. Every output carries *human_review_required* = *true* and a notice that a field officer must review it.

Data and reproducibility

Personal seed rule. LLLL = 3862. The generator uses `numpy.random.default_rng(3862)`. Base rows = 240 + (3862 mod 41) = 240 + 8 = **248**. m = round(0.05 x 248) = **12** missing values injected into each of soil_moisture_pct and fertilizer_kg, then 3 exact duplicates were appended: **251 raw rows**.

Column	Generation rule	Role
rainfall_mm	Uniform 600-1400	Feature
avg_temp_c	Uniform 12-26	Feature
soil_moisture_pct	Uniform 20-80 (12 set missing)	Feature
fertilizer_kg	Uniform 0-250 (12 set missing)	Feature
field_size_ha	Uniform 0.2-4.0	Feature
season	Random A / B	Categorical feature
cooperative_support	Random 0 / 1	Binary feature
yield_tons	Prescribed formula, noise SD 0.80, clipped at 0.50	Regression target
high_risk	Yield per hectare below its median	Classification target

Data-quality evidence

Check	Result
Raw rows / rows after removing duplicates	251 / 248 (3 duplicates removed)
Missing values in raw file	soil_moisture_pct 13, fertilizer_kg 13
Missing values after de-duplication	soil_moisture_pct 12, fertilizer_kg 12
Schema, range and leakage checks	passed (see artifacts/data_validation.json)

The raw count is 13 rather than 12 because one appended duplicate copied a row whose value had already been set to missing; removing duplicates restores 12 per column.

Fingerprint

File AI_A2_2423862.csv, saved with float_format="%.4f". SHA-256 of the final CSV bytes:
`f3a4210b542054016e2690d417ad4c5a7c89c15c82323fdc2d31503a8f3fc9bd`

Running `run_all.py` twice reproduced the same fingerprint and identical metric files (also checked by an automated test). run_all.py rejects any seed that is not 3862.

Pipeline design

Flow: generate CSV → validate (columns, ranges, missing, duplicates, leakage) → drop exact duplicates → fixed split 248 rows into train/test (test 25%, random_state=3862, stratified on high_risk; 186 train / 62 test) → fit models → evaluate on the held-out test set → save models and artifacts.

Preprocessing is a scikit-learn ColumnTransformer inside every Pipeline: numeric features use median imputation then StandardScaler; season uses most-frequent imputation then one-hot encoding (unknown categories ignored). Because imputation, scaling and encoding are steps of the Pipeline, they are fitted only on training data (and on training folds during cross-validation), and the very same fitted transformations are applied at inference time by predict.py and app.py.

Leakage controls. yield_tons, yield per hectare and high_risk are never classification inputs (an assertion and a test enforce this). high_risk is derived from yield, so a model given yield could trivially recreate it. Clustering uses the seven features only; yield and risk appear in profiles only after clustering, to interpret them.

Model selection. Candidates were compared with 5-fold cross-validation on the training split only; the best non-baseline model was chosen and the held-out test set was used once for reporting. Every random state is 3862.

Task	Baseline	Candidates	Selected (by CV)
Regression	Mean predictor	Ridge, Random Forest, Gradient Boosting	ridge_regression
Classification	Majority class	Logistic Regression, Random Forest, Gradient Boosting	logistic_regression
Clustering	-	K-Means, K = 2..6, scaled features	K = 2 (silhouette)

Saved artifacts: models/regressor.joblib, classifier.joblib and clusterer.joblib are full pipelines (preprocessing + model), plus model_metadata.json. predict.py and app.py load exactly these files.

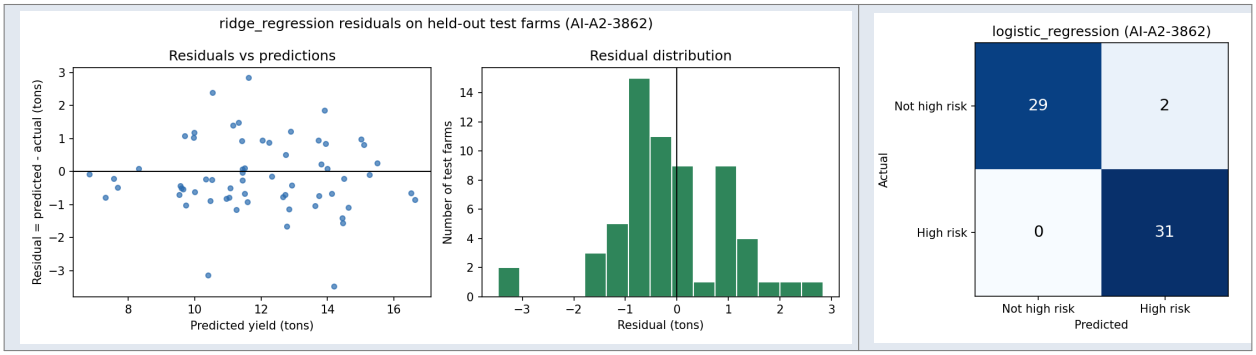
Regression and classification

Regression (held-out test set, tons)

Model	MAE	RMSE	R2	CV RMSE (train)
mean_baseline	2.068	2.538	-0.001	2.731
ridge_regression	0.867	1.117	0.806	1.043
random_forest	1.025	1.313	0.732	1.327
gradient_boosting	0.965	1.188	0.781	1.241

Classification (held-out test set)

Model	Precision	Recall	F1	ROC-AUC	CV F1 (train)
majority_baseline	0.000	0.000	0.000	0.500	0.262
logistic_regression	0.939	1.000	0.969	1.000	0.913
random_forest	0.906	0.935	0.921	0.972	0.882
gradient_boosting	0.853	0.935	0.892	0.976	0.872



Selected models. ridge_regression has the lowest 5-fold cross-validated RMSE on the training split (1.043 t vs baseline 2.731 t). The test set was not used for selection; its held-out RMSE is 1.117 t and R2 is 0.806. logistic_regression has the highest 5-fold cross-validated F1 on the training split (0.913). Held-out test F1 is 0.969, precision 0.939, recall 1.000, ROC-AUC 1.000. The test set was not used for selection.

Reading the plots. Residuals are close to zero on average (mean -0.15 t, a slight under-prediction) with no strong pattern, and the test RMSE of 1.12 t is close to the 0.80 t noise built into the generator, which no model can remove. Ridge regression won because the true yield formula is almost linear; the confusion matrix [[29, 2], [0, 31]] shows no missed high-risk farm (recall 1.00) and 2 false alarms. With only 62 test farms, an ROC-AUC of 1.000 should be read cautiously.

One important regression error. Test farm row 194: actual 17.67 t, predicted 14.20 t (error -3.47 t, about 4.3 noise standard deviations). Its fertilizer value was missing, so the model used the training median instead of the real amount, and the farm has a large field (3.59 ha), where yield is most sensitive to errors. I conclude the under-prediction comes from this imputation combined with a large positive random noise draw; I cannot separate the two because noise is unobservable.

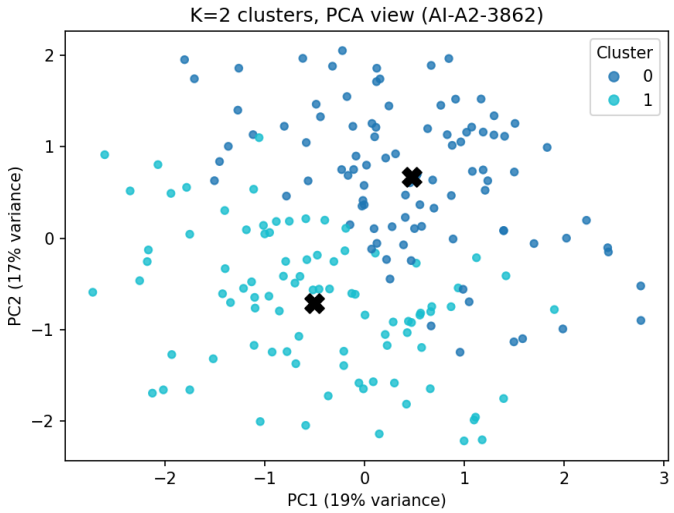
One important classification error. Farm row 35 was predicted high risk (probability 0.61) but is not. Its true yield per hectare is 6.05, only 0.42 from the median 5.63 that defines the label. Because the label is a median split, many farms sit near the boundary and a small amount of noise flips them; this is a borderline case, not a model failure.

Clustering

Features were imputed and scaled (training data only), no targets were used, and K-Means was run for K = 2..6 with n_init=10, random_state=3862.

K	2	3	4	5	6
Silhouette	0.158	0.129	0.136	0.136	0.142

K=2 has the highest silhouette score (0.158). The silhouette is low, which means the clusters overlap: farm features are independent random draws, so the data has no natural groups.



Farm profiles

Cluster	Farms (train)	Profile (plain language)	Mean yield (t)*	High-risk share*
0	96	Cluster 0: mostly without cooperative support.	11.88	0.52
1	90	Cluster 1: mostly supported by the cooperative.	12.11	0.48

*Computed after clustering only to help interpretation; never used to form clusters.

Operational meaning (one cluster). Cluster 0: mostly without cooperative support. Mean yield is 11.88 t and 52% of its farms are high risk. A field officer can use the groups to plan visits, for example one outreach message for supported farms and another for unsupported ones. Because the split is mainly driven by one binary feature and the silhouette is weak, it is a convenient grouping, not evidence of two distinct farm types.

API and testing

Endpoint	Purpose	Success	Errors
GET /health	Liveness and verification code	200 {status, verification_code}	503 if models missing
GET /model-info	Selected models, fingerprint, test metrics	200 JSON	503 if models missing
POST /predict	Yield, risk, probability, cluster	200 + verification code	422 invalid input; 503 models missing

Request validation uses a typed Pydantic model: floats with ge/le range limits, season as Literal['A','B'], cooperative_support as Literal[0,1], unknown fields forbidden.

Valid request and response

```
POST /predict {"rainfall_mm":1000,"avg_temp_c":18.5,"soil_moisture_pct":55,"fertilizer_kg":120,"field_size_ha":1.5,"season":"A","cooperative_support":1}
→ 200 {"predicted_yield_tons":11.7842,"high_risk_prediction":0,"high_risk_probability":0.1009,"cluster":1,"verification_code":"AI-A2-3862","human_review_required":true,...}
```

Invalid request (rainfall 5000, season 'C', avg_temp_c 'hot', field_size_ha missing)

```
→ 422 with one message per problem, for example: "rainfall_mm=5000 is outside the allowed range [600.0, 1400.0]"; the CLI prints the same list and exits with code 2 (no stack trace).
```

Automated tests (pytest, tests/test_system.py)

Test	What it exercises
valid inference	all output fields, verification code, human-review flag
missing field	error names field_size_ha
wrong type	text for a number; boolean for a binary field
range error	temperature 40 and season 'C' rejected
wrong seed	seed that is not 3862 is rejected
dataset signature	248 base rows, 3 duplicates, missing values, min yield 0.5
no leakage	forbidden columns are not features
reproducibility	two full trainings give same SHA-256 and metrics
API	/health, /model-info, valid /predict, 422 for bad range

Failure behaviour: invalid input never reaches the model; missing model files return a clear 503/exit code 3 asking the user to run run_all.py. The full pytest log is in evidence/AI_A2_2423862_TEST_LOG.pdf.

Model card and reflection

Item	Statement
Intended use	Classroom decision-support prototype to help a field officer prioritise visits for synthetic Musanze potato farms.
Prohibited use	Replacing an agronomist; deciding loans, insurance or public services; use on real farms without re-training and validation.
Data	100% synthetic, generated from seed 3862; relationships are the ones I coded, so scores are optimistic compared with real farms.
Limitations	Small data (248 farms, 62 test); inputs limited to the training ranges; temperature effect is simple; clusters are weak (silhouette 0.16); no pests, disease, prices or soil type.
Risks	Over-trusting a number; unfair treatment of farms without cooperative support; silent errors when a value is imputed.
Human oversight	Every output flags human_review_required; a field officer decides and can override.

Reflection

The most useful lesson was that the simple, interpretable models won: ridge regression and logistic regression beat the forests because the data were generated from near-linear rules, and cross-validation on the training split picked them without touching the test set. The two analysed errors show different limits: one caused by imputing a missing input, one by a median-based label that is unstable near its boundary.

With real data I would collect actual cooperative records, check for spatial and seasonal drift, add features such as soil type and variety, calibrate the probabilities, replace the median-based risk label with an agronomic threshold, track predictions against outcomes, and test fairness across farm sizes and support status before any operational use.